*updated: 4/26*

# Curriculum Vitae: Chad McKell

## ABOUT


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| Position | Research Engineer / Ph.D. Candidate |
| Company | Sigma Connectivity, on assignment at Meta |
| Address | 10301 Willows Road NE, Redmond, WA 98052 |
| Email | `cmckell@meta.com` |
| Website | `chadmckell.com` |
| Languages | English (native), Spanish (B2), French (A2) |
| Code | Python, C++/CUDA, MATLAB, LaTeX |


Research — My research spans mathematical modeling and numerical simulation of physical systems, with an emphasis on generating acoustic wave simulation data for computational audio and architectural acoustics applications. Recent projects include conformal symmetry for exterior wave problems; differential forms for non-spherical harmonic decomposition; and variational principles for elastic wave modeling. Past projects include underwater acoustic sensing, simulation for audio-haptics, and optical standing-wave trapping.

## EDUCATION

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|---|---|
| 9/19–(exp. 9/26) | **University of California San Diego**, Ph.D. in Computer Music<br>Research area: Computational Acoustics and Applied Mathematics<br>Dissertation: *Conformal Symmetry for Exterior Wave Problems*<br>Co-advisors: Albert Chern (CSE), Miller Puckette (Music; IRCAM) |
| 9/16–10/17 | **University of Edinburgh**, M.S. in Acoustics and Music Technology |
| 8/09–12/15 | **Wake Forest University**, M.S. in Physics |
| 6/02–8/09 | **Brigham Young University**, B.S. in Biophysics |

## EMPLOYMENT

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|---|---|
| 9/24– | **Sigma Connectivity, on assignment at Meta**, Research Engineer |
| 3/24–6/24 | **University of California San Diego**, Teaching Assistant in Computer Science |
| 6/23–2/24 | **Meta, Reality Labs Research**, Research Scientist Intern |
| 8/21–3/22 | **Meta, Reality Labs Research**, Research Intern |
| 9/19–6/23 | **University of California San Diego**, Teaching Assistant/Researcher in Music |
| 7/18–7/19 | **Applied Research in Acoustics**, R&D Scientist |
| 5/18–5/18 | **Moog Music**, Audio Software Consultant |
| 4/17–9/17 | **Lofelt**, Acoustics Research Consultant – Acquired by Meta in 2022 |
| 10/14–8/16 | **J.P. Morgan/Neovest (via ConsultNet)**, Software Development Engineer in Test |
| 8/12–12/12 | **University of North Carolina School of the Arts**, Adjunct Instructor of Physics |
| 9/09–9/11 | **Wake Forest University**, Teaching Assistant in Physics |
| 9/08–6/09 | **Brigham Young University**, Tutorial Lab Assistant in Physics |
| 8/07–3/09 | **Brigham Young University**, Research Assistant in Philosophy |

## RESEARCH ACTIVITIES

9/24– **Sigma Connectivity, on assignment at Meta**, Research Engineer
Redmond, Washington. Affiliation: Reality Labs Research Audio. Research area: *spatial audio*. Build novel spatial audio technology for Reality Labs Research at Meta. Supervisor: Sebastian Prepeliță. Team Lead: Ravish Mehra.

## RESEARCH ACTIVITIES CONT.

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|---|---|
| 9/19–9/24 | **University of California San Diego**, Ph.D. Student<br>La Jolla, California. Affiliation: Center for Visual Computing; Department of Music. Research area: *computational acoustics, applied mathematics.* Developed geometric and numerical methods for solving exterior wave problems, with applications in virtual acoustics and related fields. Committee: Albert Chern (co-chair, CSE), Miller Puckette (co-chair, Music; IRCAM), Melvin Leok (Mathematics), Stefan Bilbao (IRCAM), Sebastian Prepeliță (Meta), Shahrokh Yadegari (Music). |
| 6/23–2/24 | **Meta, Reality Labs Research**, Research Scientist Intern<br>Redmond, Washington. Affiliation: Reality Labs Research Audio. Research area: *spatial audio.* Built novel spatial audio technology for Reality Labs Research at Meta. Supervisor: Sebastian Prepeliță. Team Lead: Ravish Mehra. |
| 8/21–3/22 | **Meta, Reality Labs Research**, Research Intern<br>La Jolla, California. Research description: see above. |
| 7/18–7/19 | **Applied Research in Acoustics**, R&D Scientist<br>Culpeper, Virginia. Research area: *underwater acoustic sensing, acoustic beamforming.* Developed physics-based signal processing algorithms for object detection and classification in naval sonar systems. Team Lead: Jonathan Botts. |
| 1/17–8/17 | **University of Edinburgh**, Master's Student<br>Edinburgh, Scotland. Affiliation: Acoustics and Audio Group, Reid School of Music. Research area: *computational acoustics, numerical simulation.* Developed numerical simulations for commercial audio-haptic devices, including the Razer Nari Ultimate headphones. Received research funding from Lofelt, a Berlin-based haptics startup acquired by Meta in 2022. Advisor: Stefan Bilbao. |
| 1/10–9/13 | **Wake Forest University**, Master's Student<br>Winston-Salem, North Carolina. Affiliation: Department of Physics. Research area: *optical trapping, laser beam characterization, fluid dynamics.* Designed, constructed, and analyzed standing-wave optical traps for Brownian particle tracking. Published work in the Journal of the Optical Society of America B. Advisor: Keith Bonin. |
| 8/07–8/09 | **Brigham Young University**, Undergraduate Student<br>Provo, Utah. Affiliation: Department of Physiology and Developmental Biology. Research area: *biophysics, structural biology.* Investigated structural effects of isoflurane on artificial lipid bilayers using atomic force microscopy. Advisor: David Busath. |

## TEACHING EXPERIENCE

### ***As Instructor***

#### **UNCSA**

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| SCI 1100 | General Physics. Fall 2012 (1 term). |

**TEACHING EXPERIENCE CONT.**

***As TA***

**UCSD**

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| CSE 291 | Physics Simulation. Spring 2024 (1 term). |
| MUS 5 | Sound in Time. Spring 2020 (1 term). |
| MUS 6 | Electronic Music. Fall 2020 (1 term). |
| MUS 15 | Popular Music: David Bowie. Winter 2021 (1 term). |
| MUS 15 | Popular Music: Video Game Music. Winter 2020 (1 term). |
| MUS 171 | Computer Music I. Winter 2022 (1 term). |
| MUS 172 | Computer Music II. 2021–2022 (2 terms). |

**WFU**

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|---|---|
| PHY 113 | General Physics I (Mechanics). 2009–2011 (4 terms). |
| PHY 114 | General Physics II (E&M). Fall 2010 (1 term). |

***As Tutor***

**BYU**

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|---|---|
| PHSCS 105 | General Physics 1 (Mechanics). 2008–2009 (2 terms). |
| PHSCS 106 | General Physics 2 (E&M). Winter 2009 (1 term). |
| PHSCS 121 | Principles of Physics 1 (Mechanics). 2008–2009 (2 terms). |
| PHSCS 123 | Principles of Physics 2 (Waves/Thermo). W/Sp 2009 (2 terms). |
| PHSCS 220 | Principles of Physics 3 (E&M). W/Sp 2009 (2 terms) |

**PUBLICATIONS**

**Manuscripts in Progress**

(1) **C. McKell**, M. Nabizadeh, S. Wang, and A. Chern, "Wave simulations in infinite spacetime".

*Simulating wave propagation on an infinite domain has been a long-standing computational challenge. Conventional approaches to this problem only produce wave simulations on a small subset of the infinite domain. Using the fact that wave propagation on an infinite Minkowski spacetime is equivalent to wave propagation on a bounded Minkowski spacetime under a Kelvin-like transformation, we simulate wave propagation on the entire infinite domain using a finite discretization of the bounded domain with no additional loss of accuracy from the transformation.*

(2) **C. McKell**, *Conformal Symmetry for Exterior Wave Problems*, Ph.D. Dissertation, University of California San Diego. Defense planned for Fall 2025. Advisors: Albert Chern and Miller Puckette.

*This work presents novel geometric methods for handling open exterior boundaries in wave simulations. First, I discuss extensions to the reflectionless discrete perfectly matched layer for the wave and Helmholtz equations. Then, I demonstrate that the conformal invariance of the wave equation under a Kelvin transform in Minkowski spacetime allows one to convert an infinite domain problem into a bounded domain problem that can be solved using standard numerical methods with no additional loss of accuracy introduced by the transform. I conclude by discussing parallel computing strategies for the geometric boundary models, applications in architectural acoustics and binaural audio, and future work in obstacle boundary flattening.*

## PUBLICATIONS CONT.

### Journal Articles

(3) **C. McKell** and K. Bonin, "Optical corral using a standing-wave Bessel beam," *Journal of the Optical Society of America B*, Vol. 35, No. 8, 1910–1920, 2018.

### Conference Proceedings

(4) **C. McKell**, "Sonification of optically-ordered Brownian motion," In Proceedings of the International Computer Music Conference (ICMC), Utrecht, Netherlands, September 2016.

### Master's Theses

(5) **C. McKell**, *Real-Time Physical Modeling for Haptic Feedback Rendering*, Master's Thesis, University of Edinburgh, Acoustics and Audio Group, 2017. Advisor: Stefan Bilbao.

(6) **C. McKell**, *Confinement and Tracking of Brownian Particles in a Bessel Beam Standing Wave*, Master's Thesis, Wake Forest University, Department of Physics, 2015. Advisor: Keith Bonin.